MMET 363 Midterm 1 Formula Sheet

1. Internal Loading

$$\text{Factor of Safety} = \frac{\text{Failure/Yield Strength}}{\text{Applied Strength}} \text{ or } N = \frac{Sy}{\sigma}$$

$$\text{Allowable Stress } (\sigma) = \frac{\text{Failure Strength}}{\text{Design Factor}} \text{ or } \frac{Sy}{N}$$

$$\text{Required Area} = \frac{F_{\text{applied}} \times N}{S_y}$$

$$\text{Engineering Stress: } \sigma_e = \frac{\text{Force}}{\text{Area}}$$

2. Composite Calculations

Rule of Mixtures: $S_{uc} = S_{uf}V_f + \sigma' V_m$

$\epsilon_f = \frac{S_{uf}}{E_f}$ $\sigma' = E_m \epsilon_f$

V_f = Volume Fraction of Fiber
V_m = Volume Fraction of Matrix
S_uc = Ultimate Strength of composite

3. Shear Moment Diagrams

1. Order of the Lines

2. Find global equilibrium from Moment and Fy, Fx

3. Draw the shear moment chart

4. Find area of the boxes and use that as Moment diagrams

4. Thin-Walled Pressure Cylinder

1. Check for thin wall conditions: $\frac{r}{t} > 10$
2. Calculate hoop and longitudinal

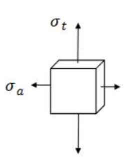
$\sigma_{\text{hoop}} = \sigma_y = \frac{Pr}{t} = \sigma_1$

$\sigma_{\text{longitudinal}} = \sigma_x = \frac{Pr}{2t} = \sigma_2$

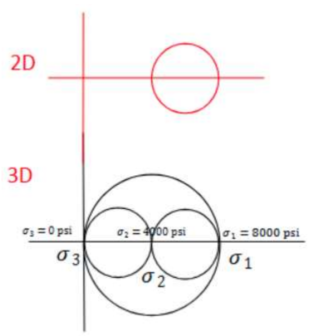
$\sigma_r = 0 = \sigma_3$

$\tau_{xy} = 0$ (usually)

3. Draw the Stress State Element:



4. Complete Mohr's Circle (Follow Instruction 5)



4. Combined Loading

1. Draw the 2D cross section with defined axes w/ sample
2. List all Fx, Fy, Fz, Mx, My, Mz vertically
3. Arrow as Axial, Bending, Shear or Torsion
4. Calculate stress components and if they're in T+ or C-

Normal Stress from **Axial** Force: $\sigma_{\text{normal (x,y,z)}} = \frac{F}{A}$ (T/C)

Normal Stress from **Bending** : $\sigma_{\text{normal (x,y,z)}} = \frac{Mc}{I}$ (T/C)

Shear Stress from **Shear** Force: $\tau_{\text{normal, direction}} = \frac{VQ}{It}$

Special Shear Eqs: $\tau_{\text{circle}} = \frac{4V}{3A}$ and $\tau_{\text{rectangle}} = \frac{3V}{2A}$

Shear Stress from **Torsional** Force: $\tau_{\text{normal, direction}} = \frac{Tc}{J}$

F = Force, A=Area, M = Moment, c = distance from neutral axis to point of interest, I = MOI, t = local thickness in the direction of shear flow, J = Polar MOI

Quick tips: Watch out for Q=0 if sample is at the top of the bar, or c=0 if bending is on Neutral Axis

5. Draw a 3D volume with matching axis. Draw all the sigma and tau forces on the volume.
6. Sum up all sigma and tau forces for Mohr Circle

5. Mohr Circle

Step 1: Find center, $o_c = \frac{\sigma_a + \sigma_b}{2}$

Step 2: Find radius, $R = \sqrt{\sigma_c^2 + \tau^2}$

Step 3: Find 2D Stresses

$\sigma_1 = \sigma_c + R$

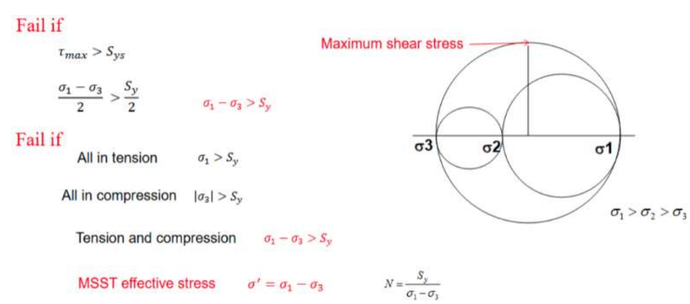
$\sigma_2 = \sigma_c - R$

Step 4: Convert to 3D Stresses

$\sigma_1 > \sigma_2 > \sigma_3$

Step 5: Draw the 3D Mohr circle w/ 2 small, 1 big circle

6. Maximum Shear Stress Theory (MSST)



1. If all in tension, safe when $N = \frac{S_y}{\sigma_1} > 1$
2. If all in compression, safe when $N = \frac{S_y}{|\sigma_3|} > 1$
3. If tension and compression, safe when $N = \frac{S_y}{\sigma_1 - \sigma_3} > 1$

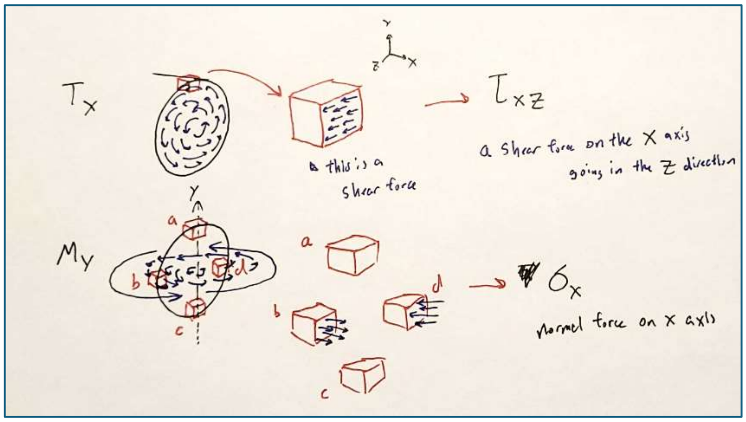
7. Distortion Energy Theory (DET)

1. Compute "Effective Stress"

$$o_e = \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}}$$

2. If $\frac{S_y}{\sigma_e} > 1$ => Safe

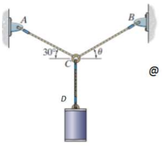
8. Miscellaneous Information



Infinitesimal Sample Notation

1. Internal Loading

The load is 1000 N and the angle θ = 35°. Determine the diameter of the wire rope (AC, BC and CD) if the yield strength of the material is 100 MPa.



$F = 1000\ N$

@ C $\sum F_x = 0 \qquad -F_{AC}\left(\frac{\sqrt{3}}{2}\right) + F_{BC}cos35° = 0$

$\sum F_y = 0 \qquad F_{AC}\left(\frac{1}{2}\right) + F_{BC}sin35° - 1000 = 0$

$F_{BC} = 955N$

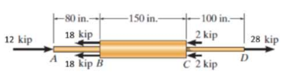
$F_{AC} = 903N$

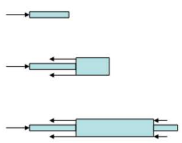
$F_{CD} = 1000N$

If use the same wire rope for all three segments, consider most critical

$$\sigma = \frac{F_{CD}}{A} < 100\ \text{MPa} \qquad A > \frac{1000\ \text{N}}{100\ \text{MPa}} = 10^{-5}\ \text{m}^2$$

The part is loaded as shown in the figure below. The cross section areas for section AB, BC, and CD are 3.0, 5.0 and 2.0 in², respectively. The yield strength of the material S_y = 60,000 psi. Identify internal loading, state tension or compression, in each section. Determine the safety factor of the design.
(1 kip = 1000 pounds-force)





Internal loadings $\sum F_x = 0$

Section

AB $\quad 12 + F_{AB} = 0 \qquad F_{AB} = -12\ (C) \qquad \sigma = \frac{F}{A} = \frac{12000}{3} = 4000$ psi

BC $\quad 12 - 18 - 18 + F_{BC} = 0 \qquad F_{BC} = 24\ (T) \qquad \sigma = \frac{F}{A} = \frac{24000}{5} = 4800$ psi

CD $\quad 12 - 18 - 18 - 2 - 2 + F_{CD} = 0 \qquad F_{CD} = 28(T) \qquad \sigma = \frac{F}{A} = \frac{28000}{2} = 14000$ psi $\qquad \boxed{N = \frac{60000}{14000} = 4.28}$

Most critical

2. Composite Calculations

A composite is made from polyester matrix and S-glass reinforcement fiber. Determine the volume fractions of the matrix and fiber so the tensile strength of the composite is **200 ksi.**

	Tensile Strength (ksi)	Elastic Modulus (ksi)	Volume Fraction
Matrix - polyester	10	400 x 10³	1-x = 68.9%
Fiber – S-glass	600	12.5 x 10³	x = 31.1%

$$\varepsilon_f = \frac{S_{uf}}{E_f} = \frac{600}{12.5 \times 10^3} = 0.048 \qquad S_{uc} = S_{uf}(x) + \sigma_m'(1-x) = \mathbf{200}$$

$$\sigma_m' = E_m \times \varepsilon_f = (400 \times 10^3)(0.048) = 19.2\ ksi \qquad S_{uc} = 600(x) + 19.2(1-x) = \mathbf{200}$$

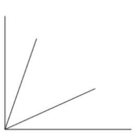
$$\mathbf{x = 0.311}$$

Compute the expected ultimate tensile strength of a composite made from unidirectional strands of carbon PAN fibers in an epoxy matrix. The volume fraction of fibers is 70%

Fiber: carbon-PAN (70%) $\quad S_{uf} = 470\ \text{ksi}, E_f = 33.5 \times 10^6$ psi

Matrix: epoxy (30%) $\quad S_{um} = 18\ \text{ksi}, E_m = 0.56 \times 10^6$ psi

$S_{uc} = S_{uf}V_f + \sigma' V_m$

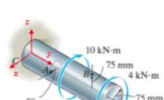


$$\varepsilon_f = \frac{S_{uf}}{E_f} = \frac{470 \times 10^3}{33.5 \times 10^6} = 0.014$$

$$\sigma' = E_m\varepsilon = (0.56 \times 10^6)(0.014) = 7840\ \text{psi}$$

$$S_{uc} = (470 \times 10^3)(0.7) + (7840)(0.3) = 331352\ \text{psi}$$

3. Torsional Shear



The solid shaft is fixed to the support at C and subjected to the torsional loadings shown. Calculate the shear stress at points A and B and sketch the shear stress on volume elements located at these points. If the shaft material has a shear yield strength of **300 MPa**, determine the factor of safety, N.

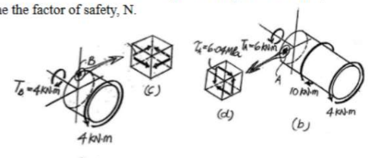
The internal torques developed at cross-sections passing through point B and A are shown in Fig. a and b, respectively.

The polar moment of inertia of the shaft is $J = \frac{\pi}{2}(0.075^4) = 49.70(10^{-6})\ \text{m}^4$. For point B, $\rho_B = C = 0.075$. Thus,

$$\tau_{xy} \qquad \tau_B = \frac{T_B c}{J} = \frac{4(10^3)(0.075)}{49.70(10^{-6})} = 6.036(10^6)\ \text{Pa} = 6.04\ \text{MPa} \qquad \text{Ans.}$$

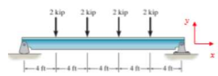
From point A, $\rho_A = 0.05$ m.

$$\tau_{yz} \qquad \tau_A = \frac{T_A \rho_A}{J} = \frac{6(10^3)(0.05)}{49.70\ (10^{-6})} = 6.036(10^6)\ \text{Pa} = 6.04\ \text{MPa}. \qquad \text{Ans.}$$

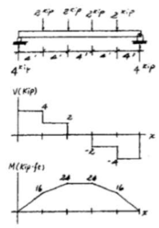


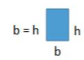
$$N = \frac{S_{ys}}{\tau} = \frac{\mathbf{300\ MPa}}{\mathbf{6.04\ MPa}} \approx \mathbf{49.69}$$

4. Shear Moment Diagrams



A beam is used to support the loads as shown. Draw the shear and the moment diagrams. If the beam has a square cross-section and the yield strength of the material is **60 ksi**. Determine the dimension of the cross section (assume **N = 3**).





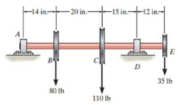
$$I = \frac{bh^3}{12} = \frac{b^4}{12}$$

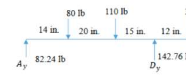
24 kip.ft = 24000(12) = 288000 lb.in

$$\sigma_x \qquad \sigma = \frac{Mc}{I} = \frac{288000(\frac{b}{2})}{\frac{b^4}{12}} \le \frac{\mathbf{60000}}{\mathbf{3}}\ \text{psi}$$

$$\frac{1728000}{b^3} \le \mathbf{20000} \qquad \mathbf{b \ge 4.42in}\ .$$

Draw the shear and moment diagrams of the beam. Determine the maximum bending moment and identify the location. Given the yield strength of the material, S_y, is 60,000 psi, calculate the required diameter to achieve the safety factor N = 2.



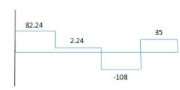


Free body diagram

The Free Body diagram

$A_y + D_y - 80 - 110 - 35 = 0$

$A_y + D_y - 225 = 0$



Shear diagram

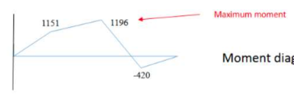
Now by applying Moment Concept :

$\sum M_A = 0$

$(80 \times 14) + (110 \times 34) - (49 \times D_y) + (35 \times 61) = 0$

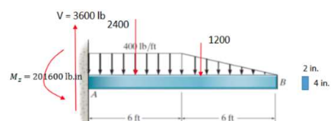
$D_y = 142.76$

$A_y = 82.24$

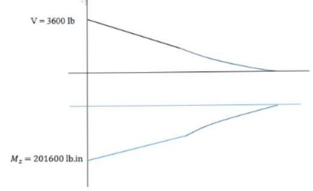


Moment diagram

Draw the shear and moment diagrams of the beam. The width and height of the cross-section of the beam are 2 in. and 4 in., respectively. Determine the required strength of material based on the stresses at location A (bottom of the beam).



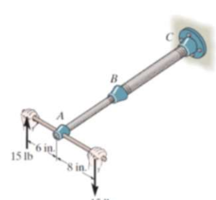
$F_z = V = 3600$



$M_z = 2400(3) + 1200(8) = 16800\ \text{lb.ft} = 201600\ \text{lb.in}$

$$\sigma = \frac{Mc}{I} = \frac{201600(2)}{\frac{2(4^3)}{12}} = 37800\ \text{psi} \le S_y$$

The assembly consists of two sections of galvanized steel bar connected using a reducing coupling at B. The smaller bar has a **diameter of 0.75 in**., whereas the larger bar has a **diameter of 1.0 in**. If the bar is tightly secured to the wall at C, determine the maximum shear stress in each section of the bar when the couple is applied to the handles of the wrench.



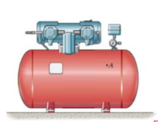
$T = 15(6 + 8) = 210\ lb$

Section AB $\qquad \tau = \frac{Tc}{J} = \frac{210\left(\frac{0.75}{2}\right)}{\frac{\pi(0.75)^4}{32}} = 2.53\ ksi$

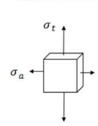
Section BC $\qquad \tau = \frac{Tc}{J} = \frac{210\left(\frac{1}{2}\right)}{\frac{\pi(1)^4}{32}} = 1.07\ ksi$

5. Thin-Walled Pressure Vessel + Mohr

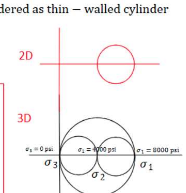
The tank of the air compressor is subjected to an internal pressure of 80 psi. If the internal diameter of the tank is 20 in., and the wall thickness is 0.1 in., determine the stress components acting at point A. Draw a volume element of the material at this point, and show the results on the element. Draw a Mohr's circle and determine the principal stresses at point A. Draw a Mohr's circle and determine the principal stresses at point A. Draw 3 circles and determine 3D principal stresses.



$$\frac{r}{t} = \frac{10}{0.10} = 100 > 10, \text{considered as thin} - \text{walled cylinder}$$

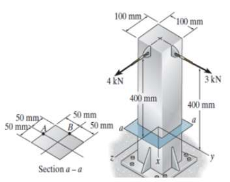


$$\boxed{\begin{array}{l} \sigma_t = \frac{pr}{t} = \frac{(80)(10)}{0.10} = 8000\ \text{psi} = \sigma_1 \\ \sigma_a = \frac{pr}{2t} = \frac{(80)(10)}{2(0.10)} = 4000\ \text{psi} = \sigma_2 \\ \sigma_r = 0 = \sigma_3 \end{array}}$$



6. Simple Combined Loading + Mohr

Given the forces acting on the object as shown. Calculate the stresses at location A. Draw a stress element of the material at this point, and show the results on the element. Draw a Mohr's circle and determine the principal stresses. Draw 3 circles and determine 3D principal stresses.



$I_y = I_z = \frac{1}{12}(0.1)(0.1^3) = 8.3333 \times 10^{-6}\, m^4$

$\Sigma F_x = 0;\ N_x = 0$ kN

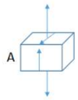
$\Sigma F_y = 0;\ V_y + 3 = 0;\ V_y = -3$ kN —Shear Force→ $\tau_{xy} = \frac{VQ}{IT} = 0$ Outside surface

$\Sigma F_z = 0;\ V_z + 4 = 0;\ V_z = -4$kN —Shear Force→ $\tau_{xz} = \frac{VQ}{IT} = \frac{3V}{2A} = \frac{3(4000)}{2(0.1)(0.1)} = 0.6$ MPa

$\Sigma M_x = 0;\ T = 0$

$\Sigma M_y = 0;\ M_y + 4(0.4) = 0;$ $M_y = -1.6$ kN·m —Bending→ $\sigma = 0$ Neutral axis

$\Sigma M_z = 0;\ M_z - 3(0.4) = 0;$ $M_z = -1.2$ kN·m —Bending→ $\sigma_x = \frac{1.2 \times 10^3 (0.05)}{8.333 \times 10^{-6}} = 7.20$ MPa (Tension)



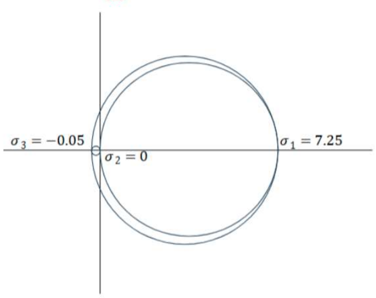
$$\sigma_c = \frac{\sigma_x + \sigma_z}{2} = \frac{7.20}{2} = 3.60$$

Note that, $\frac{\sigma_x - \sigma_z}{2} = 3.60$ and $\tau_{xz} = 0.60$

R= $\sqrt[2]{3.60^2 + 0.60^2}$ = 3.650

$\sigma_1 = \sigma_c + R = 3.60 + 3.650 = 7.250$

$\sigma_2 = \sigma_c - R = 3.60 - 3.650 = -0.05$



3D

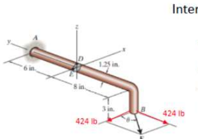
$\sigma_1 = 7.250$

$\sigma_2 = 0$

$\sigma_3 = -0.05$

7. Complex Combined Loading + Mohr + MSST

Given the force F is 600 lb, the angle θ is 45 degree, determine the internal loading at the cross section shown. Identify the stress component at location E. Draw a volume element of the material at this point, and show the results on the element. Draw a Mohr's circle and determine the principal stresses at point E. Draw 3 circles and determine 3D principal stresses. Determine the required yield strength of the material based on MSST and N = 2.



$A = \frac{\pi d^2}{4} = \frac{\pi 1.25^2}{4} = 1.23$

$J = \frac{\pi d^4}{32} = \frac{\pi 1.25^4}{32} = 0.24$

$I = \frac{\pi d^4}{64} = \frac{\pi 1.25^4}{64} = 0.12$

Internal loading / **Internal stresses @ E**

$F_x = 424$ —shear force→ $\tau_{yx} = \frac{VQ}{It} = 0$ (bottom of the section)

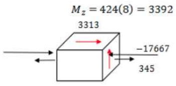
$F_y = 424$ —Axial force→ $\sigma_y = \frac{F_y}{A} = \frac{424}{1.23} = 345$ psi (tension)

$F_z = 0$

$M_x = 424(3) = 1272$ —Bending→ $\sigma_y = \frac{Mc}{I} = 0$ (neutral axis)

$M_y = 424(3) = 1272$ —Torsion→ $\tau_{yz} = \frac{Tc}{J} = \frac{1272(0.625)}{0.24} = 3313$ psi

$M_z = 424(8) = 3392$ —Bending→ $\sigma_y = \frac{Mc}{I} = \frac{3392(0.625)}{0.12} = 17667$ psi (compression)



Net $\sigma_y = -17322$ psi
$\tau_{yz} = 3313$ psi

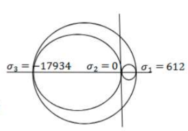
$\sigma_c = -8661$

$r = \sqrt{8661^2 + 3313^2} = 9273$

$\sigma_1 = -8661 + 9273 = 612$ psi

$\sigma_2 = 0$

$\sigma_3 = -8661 - 9273 = -17934$ psi



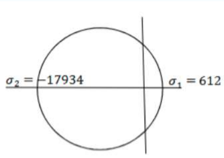
2D

$\sigma_c = -8661$

$r = \sqrt{8661^2 + 3313^2} = 9273$

$\sigma_1 = -8661 + 9273 = 612$

$\sigma_2 = -8661 - 9273 = -17934$

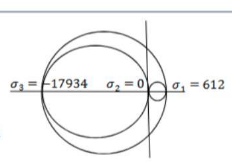


3D

$\sigma_1 = -8661 + 9273 = 612$ psi

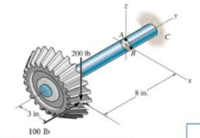
$\sigma_2 = 0$

$\sigma_3 = -8661 - 9273 = -17934$ psi



MSST $\sigma' = \sigma_1 - \sigma_3 \le \frac{S_y}{N}$ $\sigma' = 612 - (-17934) \le \frac{S_y}{2}$ $S_y \ge 37092$ psi

8. Complex Combined Loading + Mohr + DET



$D = 1.25$ in

$A = 1.23$ in^2

$I = 0.12$ in^4

$J = 0.24$ in^4

Given the loading shown, determine the internal loading ($F_x, F_y, F_z, M_x, M_y, M_z$) at the cross section passing A and B. Identify the stress component ($\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{yz}, \tau_{zx}$) at location A. Draw a stress element of the material at this point, and show the results on the element. Draw 3D Mohr's circles and determine the principal stresses ($\sigma_1, \sigma_2, \sigma_3$). Given the diameter is 1.25 in. and the material S_y = 60 ksi, determine the factor of safety, N, based on DET.

Step 1

$F_x = 0$ —normal force→

$F_y = 100$ —shear force→

$F_z = 200\ lb = V$

$M_x = 200(8) = 1600$ —bending→

$M_y = 200(3) = 600$ —torsion→

$M_z = 100(3) = 300$ —bending→

Step 2

$\sigma_y = \frac{F}{A} = \frac{100}{1.23} = 81$ (compression)

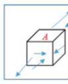
$\tau_{yz} = \frac{VQ}{It} = 0$ (on top)

$\sigma_y = \frac{Mc}{I} = \frac{1600\,(0.625)}{0.12} = 8333$ (tension)

$\tau_{yx} = \frac{Tc}{J} = \frac{(600)(1.25/2)}{0.24} = 1563$

$\sigma_? = \frac{Mc}{I} = 0$ (neutral axis)

Step 3

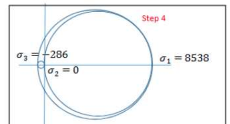


$\sigma_y = -81 + 8333 = 8252$ (tension)

$\tau_{xy} = 1563$

$\sigma_x = \sigma_z = \tau_{yz} = \tau_{zx} = 0$

Step 4



$\sigma_1 = 4126 + 4412 = 8538$

$\sigma_2 = 0$

$\sigma_3 = 4126 - 4412 = -286$

Step 5

$$\sigma_e = \sqrt{\frac{8538^2 + 286^2 + (8538 + 286)^2}{2}} = 8684 \text{ psi}$$

DET $N = \frac{S_y}{\sigma_e} = \frac{60000}{8684} = 6.9$